VED BATHINA

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EDUCATION

- **Georgia Institute of Technology** **Aug 2024 - Dec 2027**
  Bachelor of Science in Mechanical Engineering
  Relevant Coursework: Engineering Graphics and Design, Dynamics, Statics, Computing for Engineers, Computing Techniques, Creative Decisions and Design, Dynamics, Physics II, Differential Equations, Linear Algebra, Multivariable Calculus
- **Oxford University Summer Study Abroad** **May 2025 - Aug 2025**
  Completed 12 credits of interdisciplinary coursework in design, culture, and innovation across Europe and the UK

EXPERIENCE

Rover Engineer Intern | *Starpath* **May 2026 - Aug 2026**

- Managed CNC press brake tooling bring-up and operation to design and validate a repeatable in-house process for manufacturing rover thermal-system radiator panels, and built a go/no-go jig to verify panel functionality.
- Designed and manufactured headlight mounts for a lunar rover and developed a wiring harness for the system.
- Modeled, built, and tested an actuator dynamometer to verify the performance of rover actuators across varying temperatures and loads.
- Worked across the full engineering design process, from design verification matrices and NX CAD prototypes to assembly and testing.

Powertrain Engineer | *HyTech Racing Formula SAE Electric* **Dec 2025 - Present**

- Contributing to the design and construction of an electric powertrain system for a competitive Formula SAE electric racecar
- Developed knowledge of computational fluid dynamics and their application to airflow around vehicle structure and components
- Using CATIA to inspect and evaluate electric vehicle component models and assemblies including the powertrain base assembly
- Performing drivetrain assembly by clocking gears and press-fitting gearboxes onto the vehicle, ensuring proper alignment

Internals and Structures Engineer | *Georgia Tech Experimental Rocketry* **Aug 2024- Dec 2025**

- Researched, designed, and analyzed a passive staging separation system for a two-stage high-power rocket (“Live and Let Fly”)
- Modeled and constructed alignment jigs and modular rocket supports in SolidWorks; fabricated components via PLA 3D printing
- Machined aluminum staging cone using lathe, mill, and waterjet; applied GD&T and tolerance considerations for structural fit
- Designed and prototyped camera mounts (Pi Camera, RunCam Thumb Pro) for integration within nosecone avionics bay
- Designing an alignment and load-transfer system between booster and sustainer stages for 2026–27 rocket (“Mach N Roll”)

Mechanical Design Engineer | *IHOT Robotics* FIRST Robotics Competition Team **Sep 2022- May 2024**

- Designed and assembled drivetrains and mechanical subsystems for industrial-sized robots competing at the global scale
- Modeled and iterated gearbox and drivetrain assemblies in Onshape, performing fit checks and tolerance analysis
- Collaborated on the mechanical assembly and robot structure during the pre-competition build season, using metalworking tools to fabricate components while developing skills circuit building and soldering electrical connections.
- During competitions, worked in the pit crew, fixing minor and major technical issues while managing competition schedules

PROJECTS

Banzai Bill Competition Robot

- Brainstormed, modeled and fabricated a 10 lb robot to compete in a Mario themed competition involving the completion of multiple tasks including object collection and displacement
- Applied engineering drawing principles by creating isometric, orthographic, and section sketches while incorporating GD&T
- Programmed an Arduino board in C++, integrating mechatronics components such as pneumatic actuators, motors, and solenoids
- Designed rapid prototypes using SolidWorks and woodshop tools to test core subsystem mechanics prior to final assembly

Actuator Dynamometer for Lunar Rover

- Modeled, built, and tested an actuator dynamometer to verify the performance of rover actuators across varying temperatures and loads.
- Developed a python GUI using Tkinter to interface with an ADC board, thermocouple breakout boards, and ODrive motors, gaining hands-on experience with wiring, soldering, sensor operation, and motor controllers

RELEVANT SKILLS

- **Computer-Aided Design (CAD):** Autodesk Fusion 360, SolidWorks, Blender, OnShape, CATIA, NX
- **3D Printing and Prototyping (Resin and PLA)**: Bambu Studio, Bambu Lab, Formlabs, Additive and Subtractive manufacturing
- **Graphic Design:** Adobe Photoshop, Procreate, Adobe Camera Raw, Autodesk Sketchbook
- **Metal and Wood Fabrication:** Mill, Lathe, Waterjet, Laser Cutting, Woodshop machinery, CNC press brake
- **Electronics/Mechatronics:** Soldering, circuit assembly, Arduino IDE, Pneumatics, I2C/SPI communication, ODrive
- **Programming & Data Science/Machine Learning:** MatLab, Python, Unity, C++, RStudio, Microsoft Office Suite, Tkinter